Second-Order Linear Homogeneous Equations

ANSWER KEY

Section 1: Distinct Real Roots

1 Solve the auxiliary characteristic equation for $y'' - 5y' + 6y = 0$.

a. $r = 2$ **and** $r = 3$ ✓

b. $r = -2$ and $r = -3$

c. $r = 1$ and $r = 6$

d. $r = \pm\sqrt{6}$

Characteristic equation: $r^2 - 5r + 6 = 0 \Rightarrow (r-2)(r-3) = 0 \Rightarrow r = 2, 3$.

2 What is the general solution to $y'' - 5y' + 6y = 0$?

a. $y = c_1 e^{2x} + c_2 e^{3x}$ ✓

b. $y = c_1 e^{-2x} + c_2 e^{-3x}$

c. $y = (c_1 + c_2 x)e^{2x}$

d. $y = c_1 \cos(2x) + c_2 \sin(3x)$

For two distinct real roots r_1, r_2, the fundamental solution set is $\{e^{r_1 x}, e^{r_2 x}\}$, yielding $y = c_1 e^{2x} + c_2 e^{3x}$.

3 Solve the IVP $y'' - y = 0$ with $y(0) = 4$ and $y'(0) = 2$.

a. $y = 3e^{x} + e^{-x}$ ✓

b. $y = 2e^{x} + 2e^{-x}$

c. $y = 4e^{x}$

d. $y = e^{x} + 3e^{-x}$

$r^2 - 1 = 0 \Rightarrow r = \pm 1 \Rightarrow y = c_1 e^{x} + c_2 e^{-x}$. $y'(x) = c_1 e^{x} - c_2 e^{-x}$. $y(0) = c_1 + c_2 = 4$ and $y'(0) = c_1 - c_2 = 2$. Adding: $2c_1 = 6 \Rightarrow c_1 = 3, c_2 = 1$. Thus $y = 3e^{x} + e^{-x}$.

Section 2: Repeated Real Roots

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4 What is the general solution to $y'' - 6y' + 9y = 0$?

a. $y = (c_1 + c_2 x)e^{3x}$ ✓

b. $y = c_1 e^{3x} + c_2 e^{-3x}$

c. $y = c_1 e^{3x} + c_2 x^2 e^{3x}$

d. $y = c_1 \cos(3x) + c_2 \sin(3x)$

Characteristic equation: $r^2 - 6r + 9 = (r-3)^2 = 0 \Rightarrow r = 3$ (repeated root). The general solution is $y = (c_1 + c_2 x)e^{3x}$.

5 Why is the factor x introduced into the second solution xe^{rx} for repeated roots?

a. **To ensure linear independence from the first solution** e^{rx} ✓

b. Because the derivative of e^{rx} contains x

c. To make the solution satisfy $y(0) = 0$

d. By Euler's identity

Reduction of order proves that $y_2 = xe^{rx}$ is linearly independent of $y_1 = e^{rx}$, giving a non-zero Wronskian.

6 Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = -1$.

a. $y = (2 + 3x)e^{-2x}$ ✓

b. $y = (2 - x)e^{-2x}$

c. $y = 2e^{-2x}$

d. $y = (2 + x)e^{-2x}$

$r = -2$ repeated: $y = (c_1 + c_2 x)e^{-2x}$. $y(0) = c_1 = 2$. $y'(x) = c_2 e^{-2x} - 2(c_1 + c_2 x)e^{-2x}$. $y'(0) = c_2 - 2c_1 = -1 \Rightarrow c_2 - 4 = -1 \Rightarrow c_2 = 3$. Thus $y = (2 + 3x)e^{-2x}$.

Section 3: Complex Conjugate Roots

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7 If the characteristic equation has complex roots $r = \alpha \pm i\beta$, what is the real general solution?

a. $y = e^{\alpha x}(c_1\cos(\beta x) + c_2\sin(\beta x))$ ✓

b. $y = c_1 e^{\alpha x} + c_2 e^{\beta x}$

c. $y = e^{\beta x}(c_1\cos(\alpha x) + c_2\sin(\alpha x))$

d. $y = c_1\cos((\alpha + \beta)x)$

By Euler's formula $e^{(\alpha \pm i\beta)x} = e^{\alpha x}(\cos(\beta x) \pm i\sin(\beta x))$, the real fundamental solution basis is $\{e^{\alpha x}\cos(\beta x), e^{\alpha x}\sin(\beta x)\}$.

8 Find the general solution to $y'' + 16y = 0$.

a. $y = c_1\cos(4x) + c_2\sin(4x)$ ✓

b. $y = c_1 e^{4x} + c_2 e^{-4x}$

c. $y = (c_1 + c_2 x)e^{4x}$

d. $y = c_1\cos(16x) + c_2\sin(16x)$

$r^2 + 16 = 0 \Rightarrow r = \pm 4i$ (with $\alpha = 0, \beta = 4$). Thus $y = c_1\cos(4x) + c_2\sin(4x)$.

9 Solve the characteristic equation for $y'' - 2y' + 5y = 0$.

a. $r = 1 \pm 2i$ ✓

b. $r = 2 \pm 4i$

c. $r = -1 \pm 2i$

d. $r = 1 \pm 4i$

Quadratic formula: $r = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$.

Section 4: Harmonic Motion and Damping Regimes

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10 A mechanical oscillator modeled by $my'' + cy' + ky = 0$ is underdamped when:

a. $c^2 - 4mk < 0$ **(oscillatory motion with exponentially decaying amplitude)** ✓

b. $c^2 - 4mk = 0$ (critically damped)

c. $c^2 - 4mk > 0$ (overdamped, no oscillations)

d. $c = 0$ (undamped simple harmonic motion)

When the discriminant $c^2 - 4mk < 0$, the characteristic roots are complex conjugates, causing oscillatory motion that decays exponentially.

11 Which damping condition returns a displaced mass to equilibrium in the shortest time without oscillating?

a. **Critical damping ($c^2 = 4mk$)** ✓

b. Overdamping ($c^2 > 4mk$)

c. Underdamping

d. Zero damping

Critical damping provides the most rapid return to equilibrium without experiencing overshoot or oscillation.

12 What is the natural frequency ω_0 of the undamped oscillator $y'' + 25y = 0$?

a. $\omega_0 = 5$ rad/s ✓

b. $\omega_0 = 25$ rad/s

c. $\omega_0 = 10$ rad/s

d. $\omega_0 = 2.5$ rad/s

In $y'' + \omega_0^2 y = 0$, $\omega_0^2 = 25 \Rightarrow \omega_0 = 5$ rad/s.

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